

C LANGUAGE - LOOP EXERCISES

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Level 1: Basic Summations and Counting

1. Sum of natural numbers $\rightarrow 1 + 2 + 3 + \dots + n$
2. Sum of even numbers up to n
3. Sum of odd numbers up to n
4. Sum of squares of numbers $\rightarrow 1^2 + 2^2 + \dots + n^2$
5. Sum of cubes of numbers $\rightarrow 1^3 + 2^3 + \dots + n^3$
6. Factorial of a number ($n! = 1 \times 2 \times \dots \times n$)
7. Compute e^x using Taylor Series: $e^x = 1 + x + \frac{x^2}{2!} + \dots$

Level 2: Patterns (Using Nested Loops)

8. Print square pattern of *
9. Right-angled triangle pattern
10. Right-aligned triangle pattern
11. Pyramid pattern
12. Inverted triangle pattern
13. Number pattern 1 12 123 1234
14. Number repetition pattern 1 22 333 4444
15. Alphabet pattern A AB ABC ABCD
16. Palindromic number pyramid

Level 3: Numeric Problems

17. Fibonacci series (0, 1, 1, 2, 3, 5, ...)
18. Sum of digits of an integer
19. Reverse a number
20. Check if a number is palindrome
21. Find divisors of a number
22. Check if a number is prime
23. Count digits in a number
24. Check if a number is perfect
25. Check if a number is Armstrong (e.g. 153)

Level 4: Logical / Applied Exercises

26. Generate multiplication table for a given number
27. Password check (loop until correct password)
28. Keep summing numbers until user enters 0
29. Find smallest and largest digit in a number
30. Convert a decimal number to binary (print in reverse order)